

Day 6 — Anurūpyena — $\times 12$ to $\times 19$

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: extend the method to $\times 12$ and $\times 13$ using one-more-digit adjustments.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Move beyond $\times 11$.** Tirthaji’s *Anurūpyena* (“proportionately”) sūtra extends the digit-sum idea to $\times 12$ through $\times 19$. Procedure for $\times 12$:
 - Write any number N .
 - Compute: $N \times 12 = (N \times 10) + (N \times 2)$ — easy mental sum.
 - Equivalent digit-procedure: split-double-add.
2. **Worked example, 23×12 .**
 - $23 \times 10 = 230$
 - $23 \times 2 = 46$
 - Sum: 276
 - OR digit-wise: $2_3 \rightarrow 2_ (2 \cdot 1 + 3) _ (3 \cdot 1)$ wait — this gets tangled. The cleanest mental version is the $(\times 10) + (\times k)$ split.
3. **General $\times (10+k)$ rule.** $N \times (10+k) = 10N + kN$. The Anurūpyena framing emphasises the proportional scaling.
4. **Practice.** Compute mentally:

- $14 \times 13 = 140 + 42 = 182$
 - $25 \times 15 = 250 + 125 = 375$
 - $32 \times 17 = 320 + 224 = 544$
5. **Honest historical note.** The $(10+k)$ decomposition isn't unique to Tirthaji — Western mental-math systems describe the same trick. Tirthaji's contribution is grouping it under a Sanskrit sūtra-system.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Anurūpyena — $\times 12$ to $\times 19$.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).